

Project Output Summary

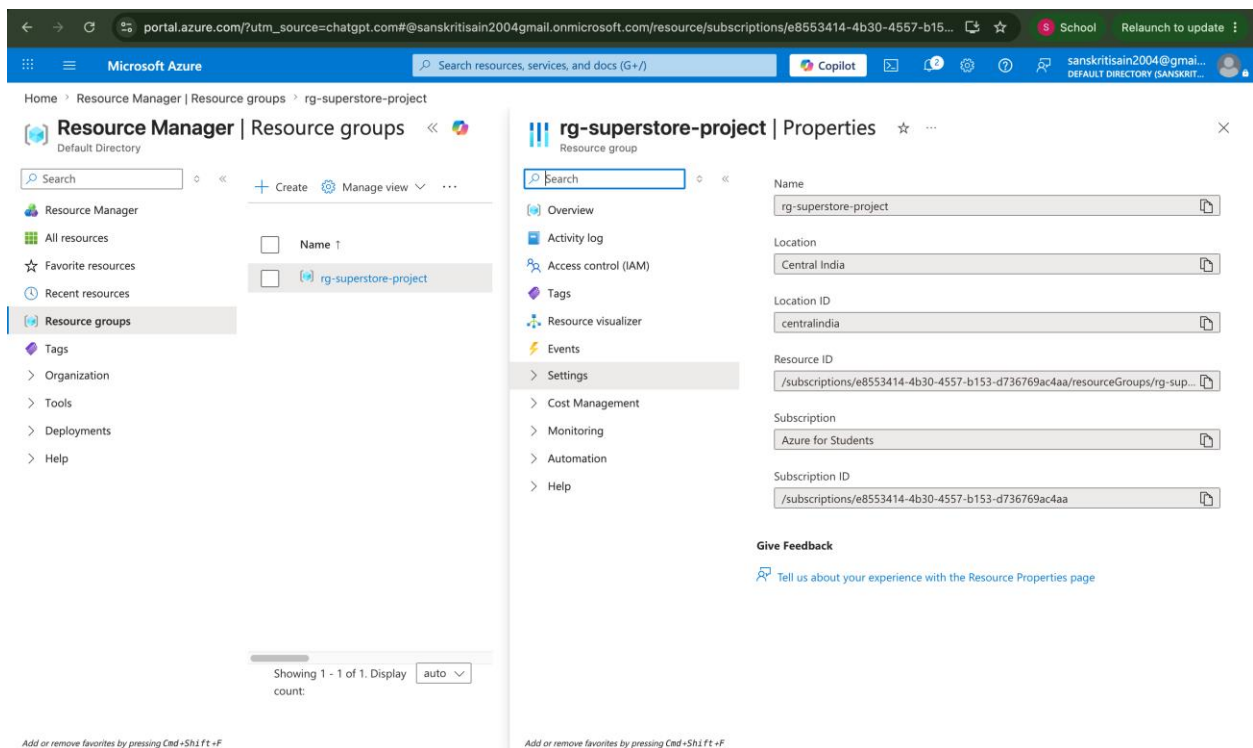
Project Title

Azure Cloud Fundamentals and Data Pipeline Implementation using Azure Data Factory

Screenshots Collected

1. Resource Group

Screenshot: 01_resource_group.png



Description:

A Resource Group named rg-superstore-project was created to organize and manage all Azure resources used in this project.

2. Storage Account

Screenshot: 02_storage_account.png

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

sanskritisain2004@gmail...
DEFAULT DIRECTORY (SANSKRIT...

Home

Deployment

Deployment is in progress

Deployment name : stsuperstore2026_1781247840407
Subscription : Azure for Students
Resource group : rg-superstore-project

Start time : 12/06/2026, 12:35:43
Correlation ID : e1fab655-0862-45cc-af2e-22abdab32ecf

Deployment details

Resource	Type	Status	Operation details
stsuperstore2026	Storage account	Accepted	Operation details

Microsoft Defender for Cloud
Secure your apps and infrastructure
[Go to Microsoft Defender for Cloud >](#)

Free Microsoft tutorials
[Start learning today >](#)

Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
[Find an Azure expert >](#)

Add or remove favorites by pressing Cmd+Shift+F

Description:

An Azure Storage Account was created to store source and destination data files used in the pipeline.

3. Blob Containers and CSV Upload

Screenshot: 03_blob_container.png

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

sanskritisain2004@gmail...
DEFAULT DIRECTORY (SANSKRIT...

Home > Storage center | Blob Storage > stsuperstore2026 | Containers

source-container

Container

source-container

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superstore.csv	12/06/2026, 18:10:07	Hot (inferred)	Block blob	2.18 MiB	Available

Add or remove favorites by pressing Cmd+Shift+F

Description:

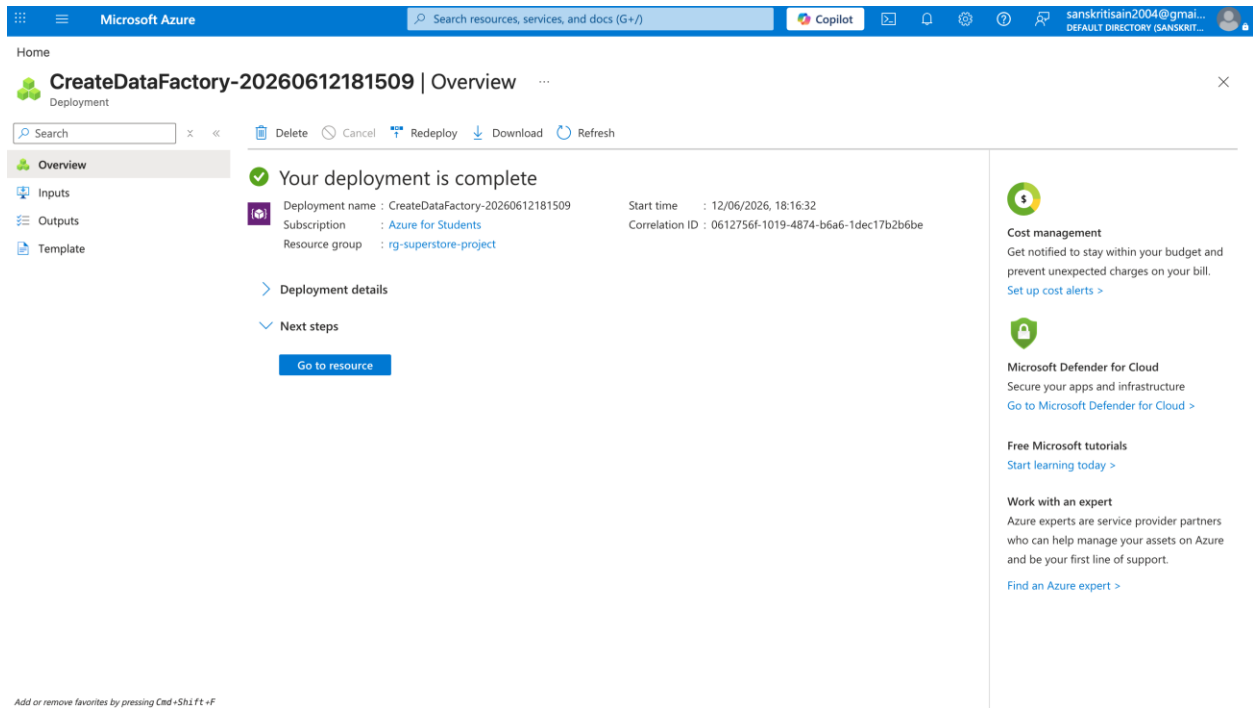
Two Blob Storage containers were created:

- source-container
- destination-container

The file Sample-Superstore.csv was uploaded successfully to the source-container.

4. Azure Data Factory Created

Screenshot 4: 04_azure_data_factory_created.png

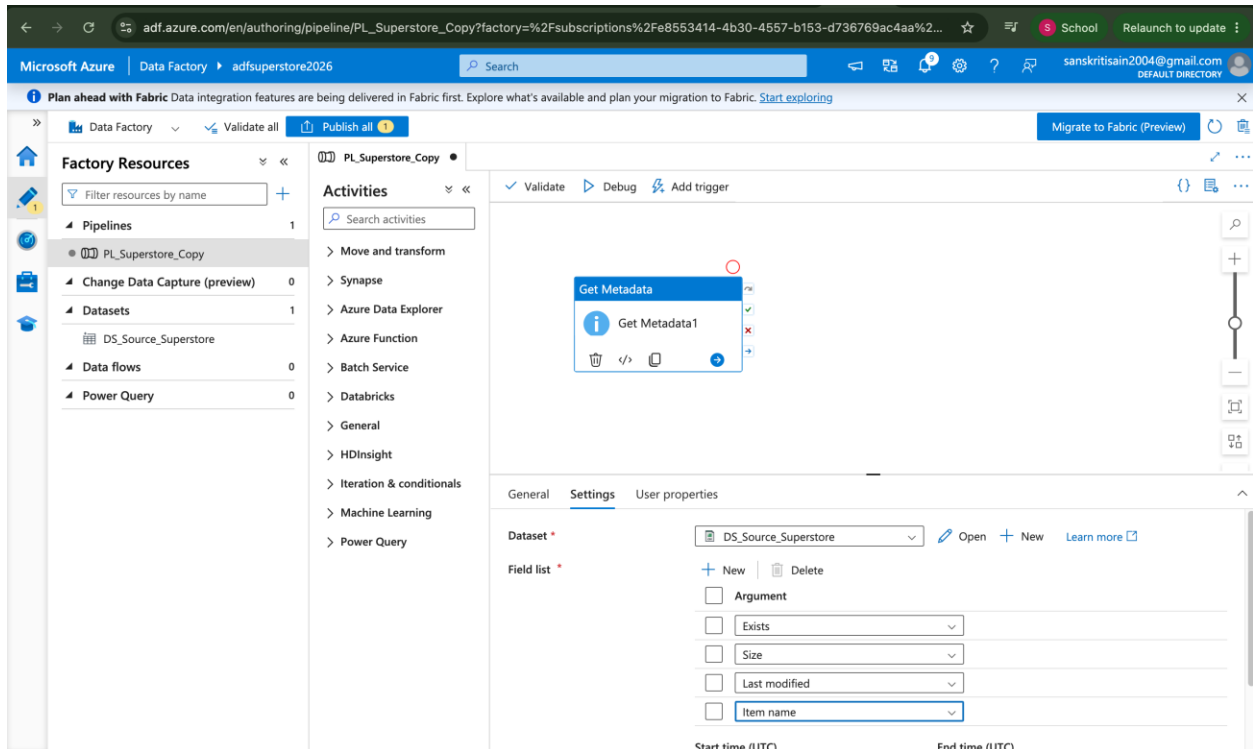


Description:

An Azure Data Factory instance was successfully created to build and manage the data pipeline.

5. Azure Data Factory

Screenshot: 05_adf_ui.png

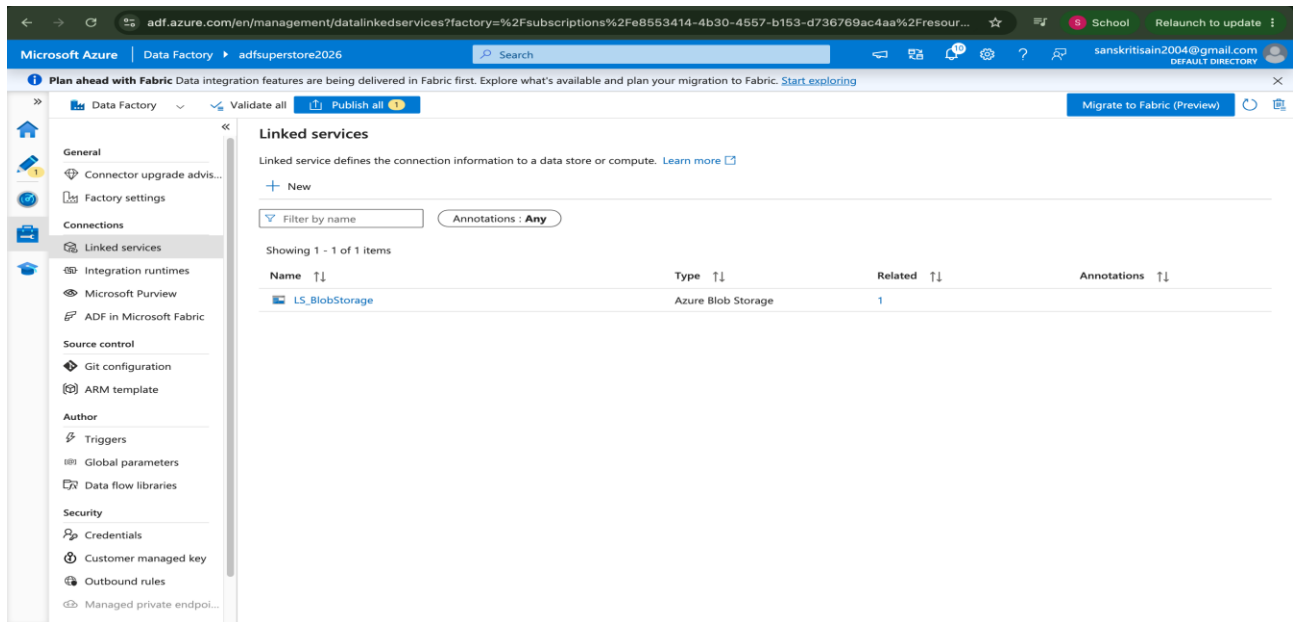


Description:

Azure Data Factory was created and explored using the Author, Monitor, and Manage modules.

6. Linked Service

Screenshot: 06_linked_service.png

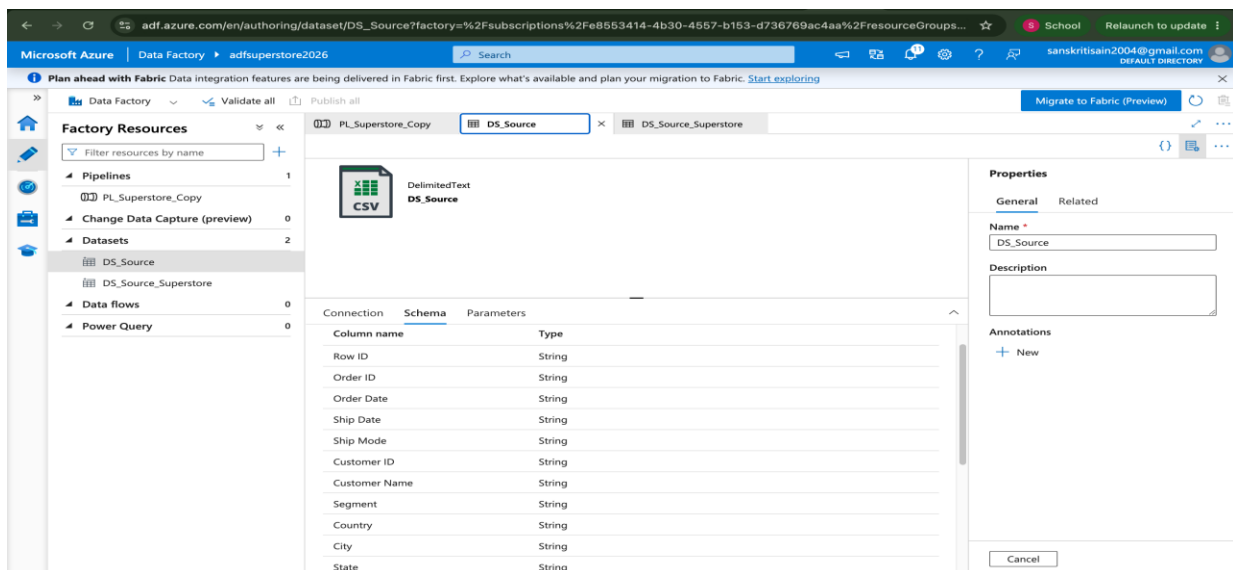


Description:

A Linked Service was created to establish a connection between Azure Data Factory and Azure Blob Storage.

7. Source Dataset

Screenshot: 07_source_dataset.png

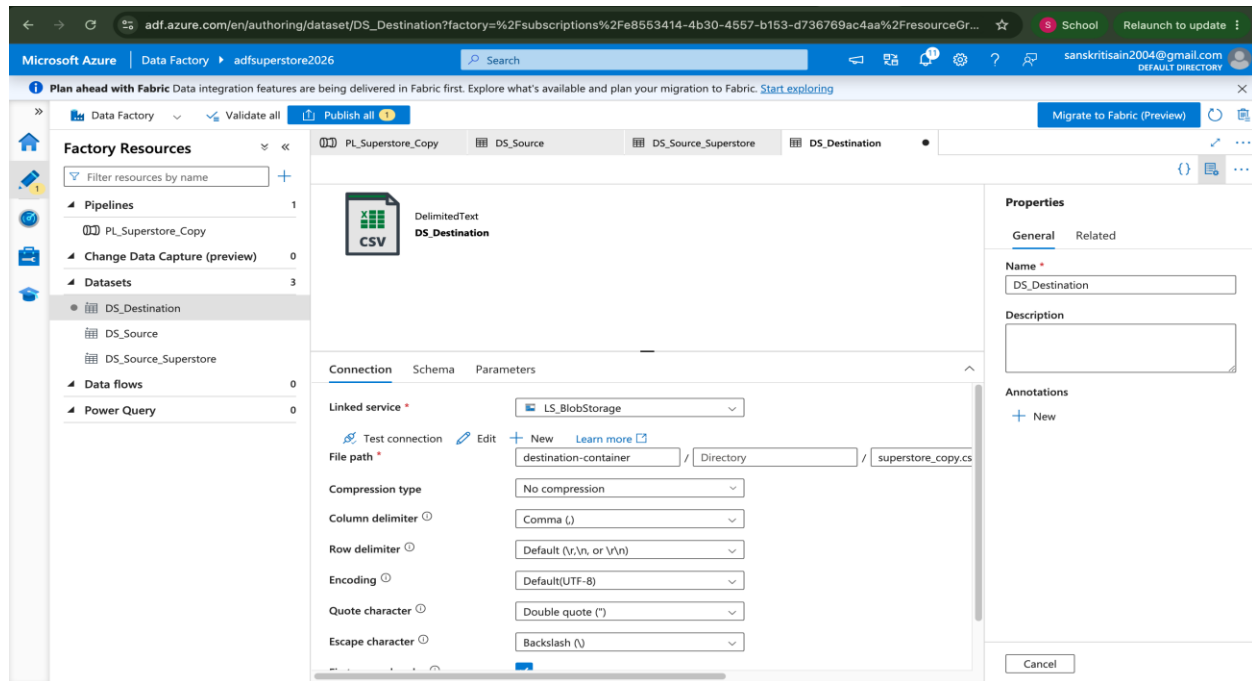


Description:

The source dataset was configured to read data from the file Sample-Superstore.csv stored in the source-container.

8. Destination Dataset

Screenshot: 08_destination_dataset.png

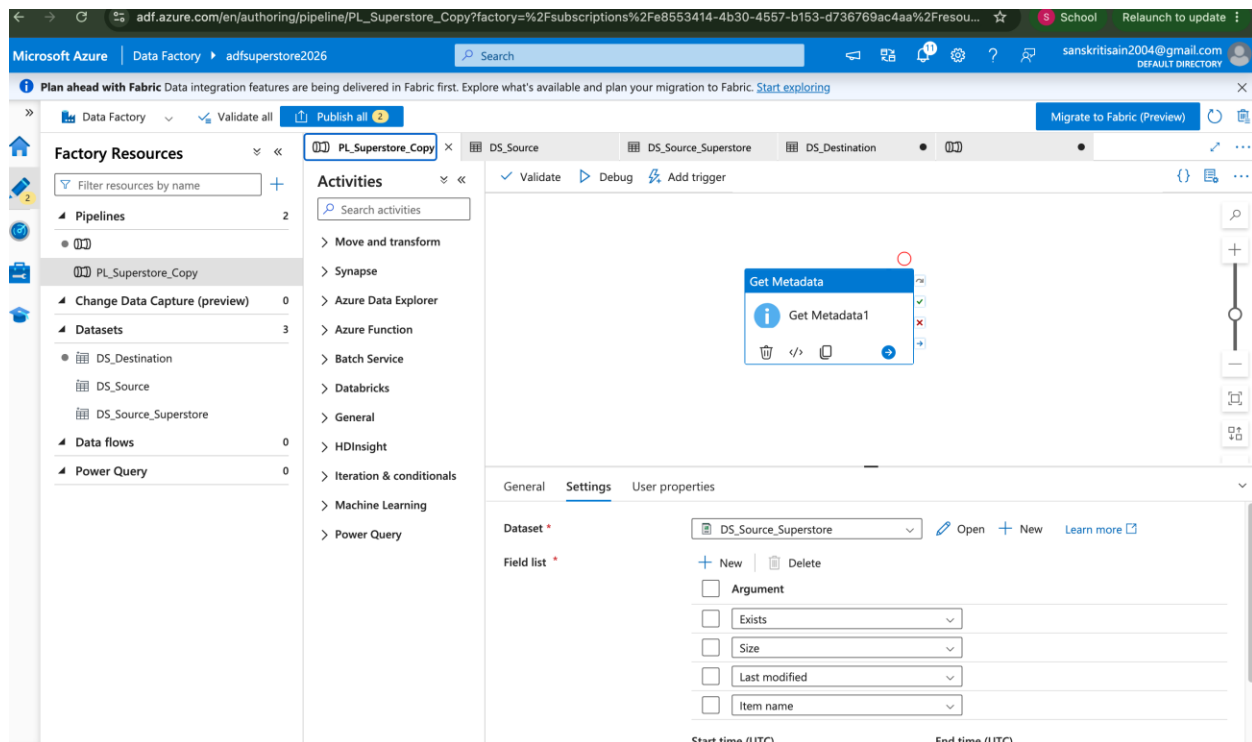


Description:

The destination dataset was configured to write data to superstore_copy.csv in the destination-container.

9. Get Metadata Activity

Screenshot: 09_get_metadata.png

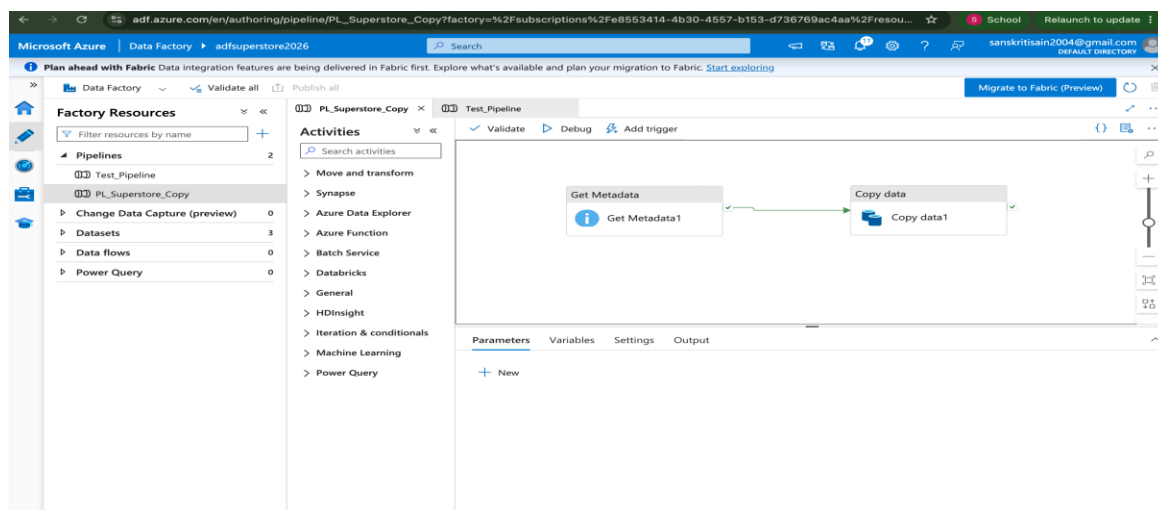


Description:

The Get Metadata activity was configured to validate the source file before processing. The activity checks file existence and metadata information.

10. Pipeline Design

Screenshot: 10_pipeline_design.png



Description:

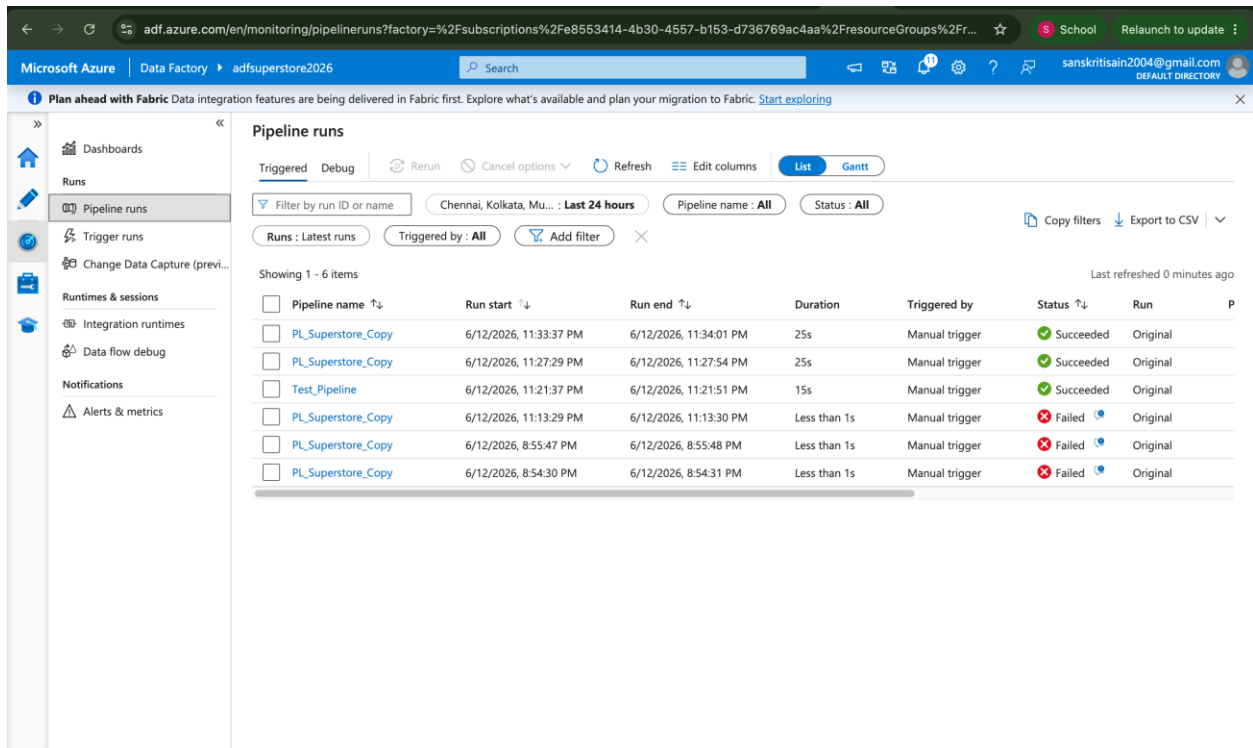
A pipeline named PL_Superstore_Copy was created using:

- Get Metadata Activity
- Copy Data Activity

The activities were connected to form a complete workflow.

11. Pipeline Execution

Screenshot: 11_pipeline_success.png



The screenshot shows the 'Pipeline runs' page in the Azure Data Factory portal. The left sidebar contains navigation options: Dashboards, Runs, Pipeline runs (selected), Trigger runs, Change Data Capture (previous), Runtimes & sessions, Integration runtimes, Data flow debug, Notifications, and Alerts & metrics. The main area displays a table of pipeline runs. The table has columns for Pipeline name, Run start, Run end, Duration, Triggered by, Status, and Run. The runs are filtered by 'Last 24 hours' and 'Pipeline name: All'. The table shows 6 items, with the first three being successful and the last three being failed.

Pipeline name	Run start	Run end	Duration	Triggered by	Status	Run
PL_Superstore_Copy	6/12/2026, 11:33:37 PM	6/12/2026, 11:34:01 PM	25s	Manual trigger	Succeeded	Original
PL_Superstore_Copy	6/12/2026, 11:27:29 PM	6/12/2026, 11:27:54 PM	25s	Manual trigger	Succeeded	Original
Test_Pipeline	6/12/2026, 11:21:37 PM	6/12/2026, 11:21:51 PM	15s	Manual trigger	Succeeded	Original
PL_Superstore_Copy	6/12/2026, 11:13:29 PM	6/12/2026, 11:13:30 PM	Less than 1s	Manual trigger	Failed	Original
PL_Superstore_Copy	6/12/2026, 8:55:47 PM	6/12/2026, 8:55:48 PM	Less than 1s	Manual trigger	Failed	Original
PL_Superstore_Copy	6/12/2026, 8:54:30 PM	6/12/2026, 8:54:31 PM	Less than 1s	Manual trigger	Failed	Original

Description:

The pipeline was executed successfully using Azure Data Factory Monitor.

Pipeline Status:

- Get Metadata → Succeeded
- Copy Data → Succeeded

12. Metadata Validation

Screenshot: 12_metadata_validation.png

adf.azure.com/en/monitoring/pipeline runs/e8eab9a3-ebc8-4440-8262-bb165ea98994?factory=%2Fsubscriptions%2F8553414-4b30-4557-b...

Microsoft Azure | Data Factory | adfsuperstore2026

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Dashboard
Runs
Pipeline runs
Trigger runs
Change Data Capture (previ...
Runtimes & sessions
Integration runtimes
Data flow debug
Notifications
Alerts & metrics

Rerun Cancel Refresh Update pipeline List Gantt

Pipeline was modified after this run. The current pipeline configuration is shown.

Get Metadata Get Metadata1 Copy data Copy data1

Output

Copy to clipboard

```
{
  "exists": true,
  "size": 2287806,
  "lastModified": "2026-06-12T12:40:07Z",
  "itemName": "Sample - Superstore.csv",
  "effectiveIntegrationRuntime":
    "AutoResolveIntegrationRuntime (Central India)",
  "executionDuration": 0,
  "durationInQueue": {}
}
```

Monitor in Azure Metrics Export to CSV

	Run start	Duration	Integration runtime	User prop...	#
Get Metadata1	6/12/2026, 11:33:44 PM	16s	AutoResolveIntegrationRuntime (Central India)		6
Get Metadata	6/12/2026, 11:33:44 PM	12s	AutoResolveIntegrationRuntime (Central India)		5

Description:

Metadata validation was completed successfully using the Get Metadata activity before the file copy process.

13. Destination File Created

Screenshot: 13_destination_file_created.png

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

sanskritisain2004@gmail...
DEFAULT DIRECTORY (SANSKRIT...

Home > Storage center | Blob Storage > stsuperstore2026 | Containers

destination-container

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

destination-container

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

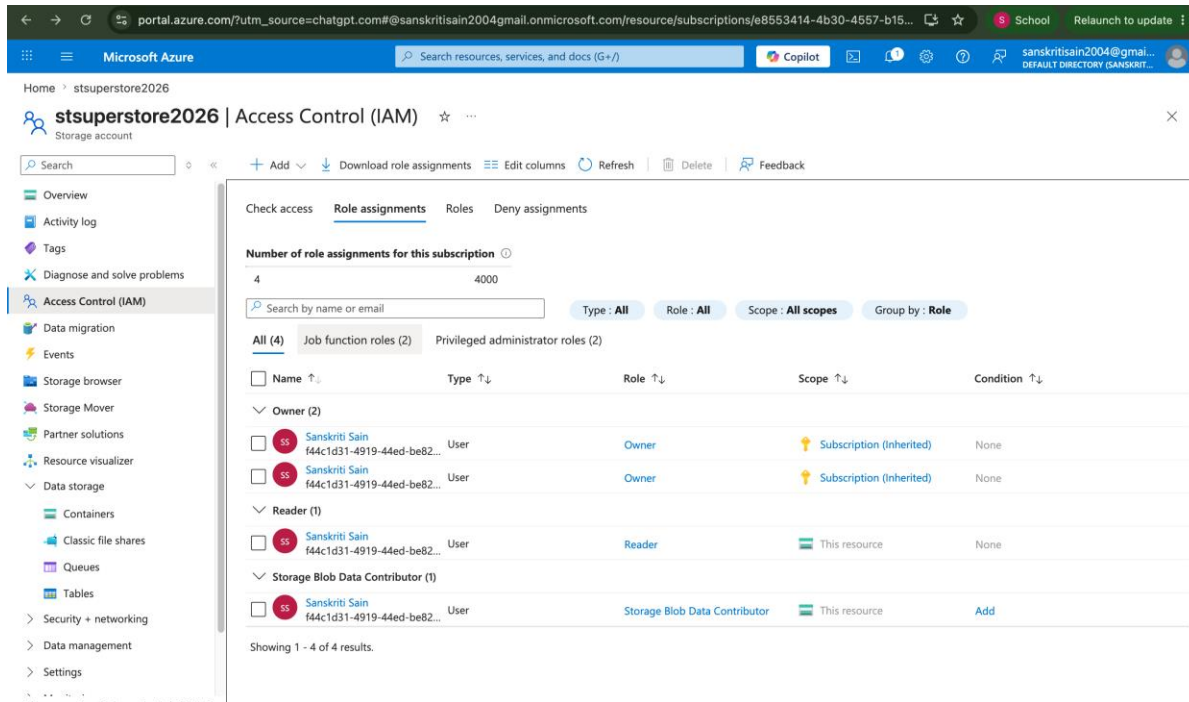
☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	superstore_copy.csv	12/06/2026, 23:27:49	Hot (Inferred)	Block blob	2.18 MiB	Available

Description:

The file superstore_copy.csv was successfully generated in the destination-container after pipeline execution.

14. IAM Role Assignment

Screenshot: 14_role_assignment.png



Description:

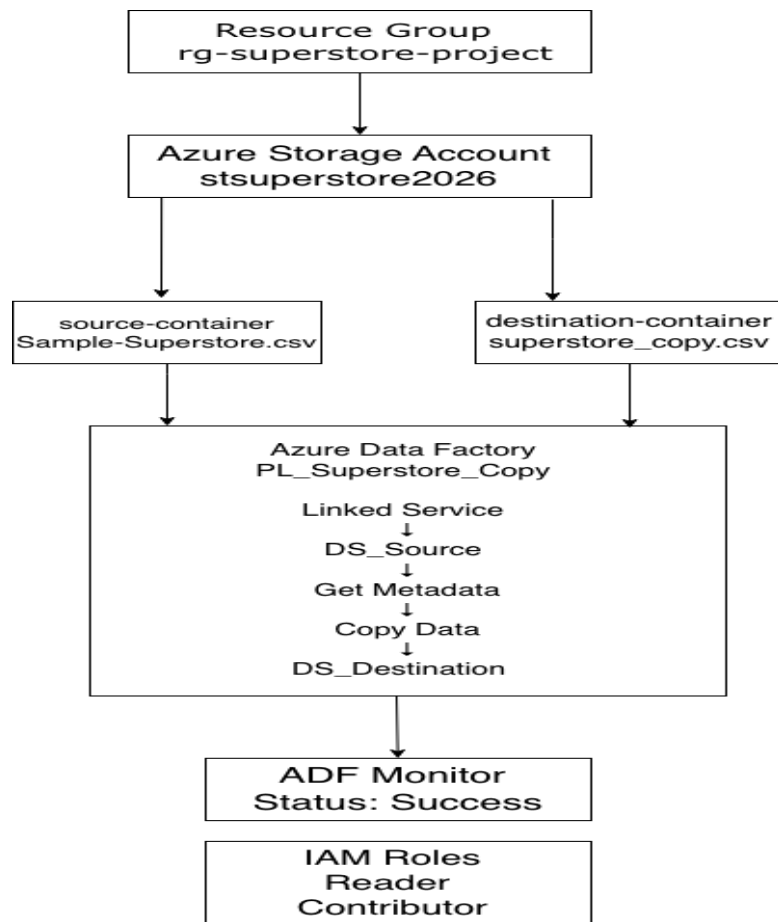
IAM permissions were configured to provide secure access to Azure resources.

Assigned Roles:

- Reader
- Contributor / Storage Blob Data Contributor

15. Architecture Diagram

Screenshot: 15_architecture_diagram.png



Description:

The architecture diagram illustrates the complete flow from Blob Storage to Azure Data Factory and finally to the destination container.

Pipeline Execution Results

Activity	Status
Get Metadata	Succeeded
Copy Data	Succeeded
Pipeline Execution	Succeeded
Metadata Validation	Successful
Data Transfer	Successful

Brief Summary

This project successfully implemented an end-to-end Azure data pipeline using Azure Blob Storage and Azure Data Factory. A source CSV file was uploaded to Azure Blob Storage and connected to Azure Data Factory through a Linked Service. The Get Metadata activity validated the source file before processing, and the Copy Data activity transferred the file from the source-container to the destination-container. Pipeline execution was monitored through Azure Data Factory Monitor, and IAM roles were configured to manage resource access securely. The project achieved all required objectives, including metadata validation, successful pipeline execution, and data transfer to the destination location.